Assignments (we're almost done!)

1. Peer edits are due Friday at 12 pm
2. Work on your talk. Talks begin Mar 13th and order will be random
3. Final papers due Mar 13t^{h} (use Canvas)

All about talks

Fish 290

Tuesday Mar 4th

First, what is a talk?

- A "talk" is an oral summary of a scientific study or studies
- Talks can be short (~10 minutes) or long (1 hour)
- Like posters, talks at conferences are a quicker way of communicating a research study than reading an entire paper

Types of talks

- Traditional presentation at scientific conferences
  - 10-15 minutes including Q&A time
  - Usually covers an entire study
- Lightning talks
  - ~3-5 minutes
  - Usually covers a single key finding or preliminary results
- Seminars
  - 30 to 60 minutes
  - Covers broad topics, can span many studies
  - Invited talks, job interviews, keynote speakers, guest lectures

Differences between talks and posters

Talks:

1. Happen once and are gone: each point must be instantly absorbed
2. Two-way communication is difficult and rushed
3. Audience is large: an auditorium full of people
4. Graphics must be simple- only displayed for a short while
5. Must maintain audience's attention the entire time, or else the message can be lost

Posters:

6. Can be read and re-read, are hung up for a long time
7. Excellent for two-way communication
8. Audience is small: one person or a few people at a time
9. Graphics can be more complex
10. People can read at their own pace, or only read parts of the poster

The biggest difference between a talk and a poster or scientific paper is: **the audience must absorb information quickly to follow along!**

Giving a good talk means understanding an audience

Everyone is listening at the beginning

People perk up at the end

People lose attention in the middle

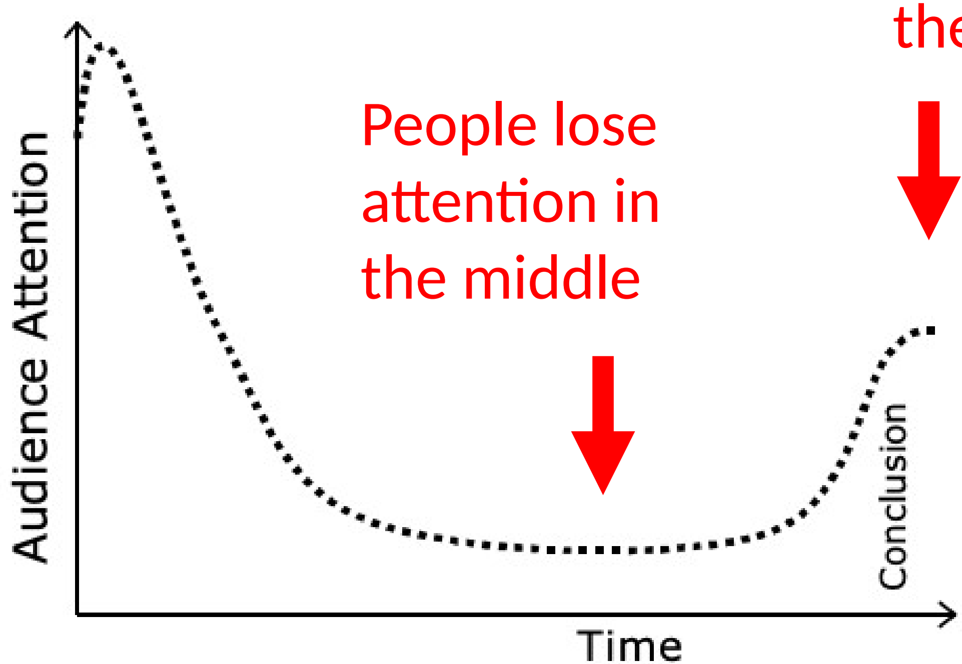


Figure 1 Typical attention the audience pays to an average presentation

Giving a good talk means understanding an audience

If we want to maximize the audience's attention, what can we do?

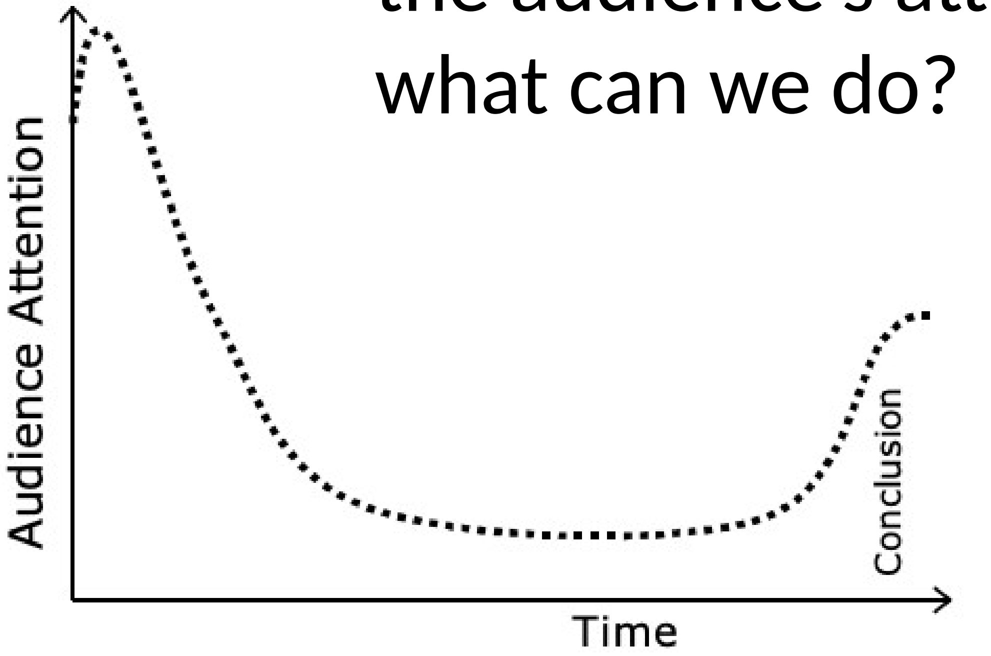


Figure 1 Typical attention the audience pays to an average presentation

Giving a good talk means understanding an audience

1. Emphasize the beginning and end

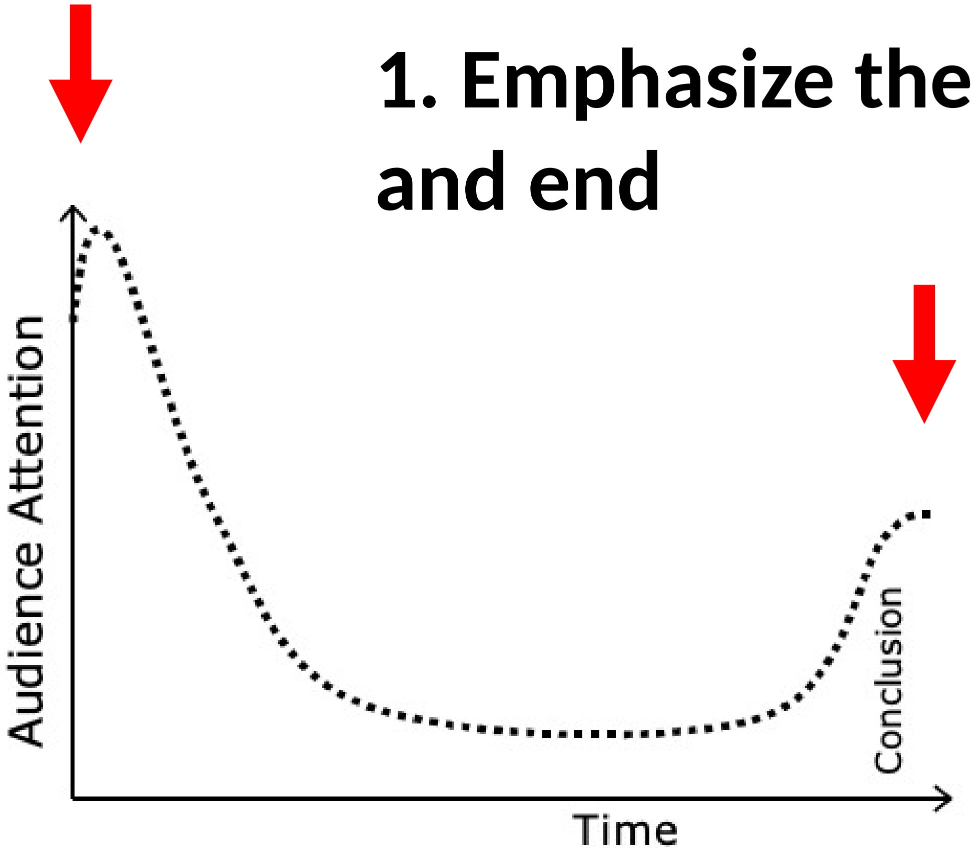


Figure 1 Typical attention the audience pays to an average presentation

Giving a good talk means understanding an audience

1. **Emphasize the beginning and end**

But what about the middle?

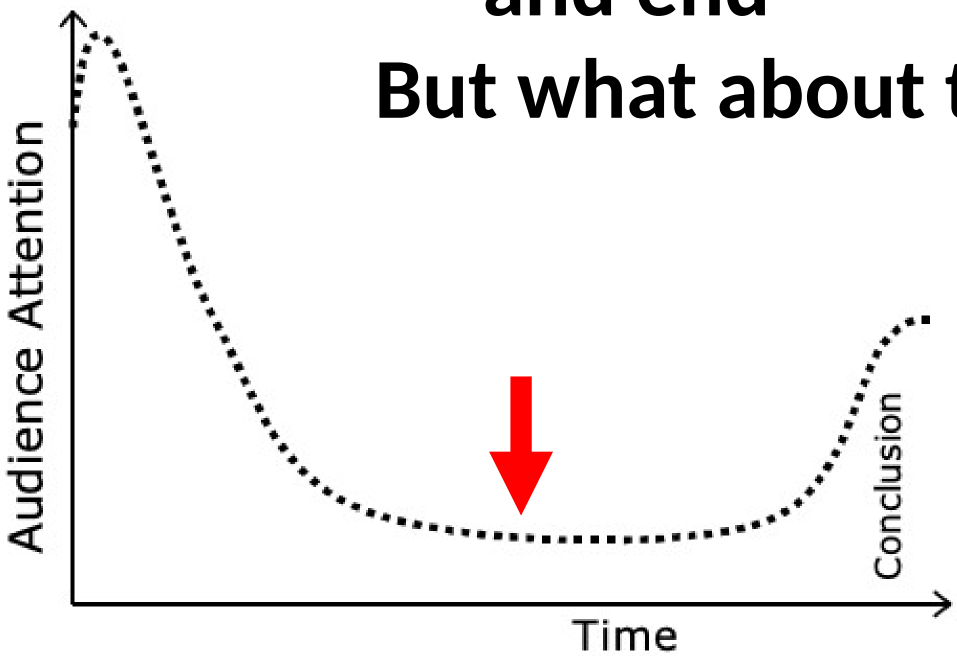


Figure 1 Typical attention the audience pays to an average presentation

Giving a good talk means understanding an audience

What makes the audience lose focus?

- **Not enough background information**
- **Structure of the presentation is hard to follow**
- **Visual aids are confusing or too fast**
- **Too much information on slides or too many slides**
- **Speaker is talking too fast**
- Other reasons you can think of?

Giving a good talk means understanding an audience

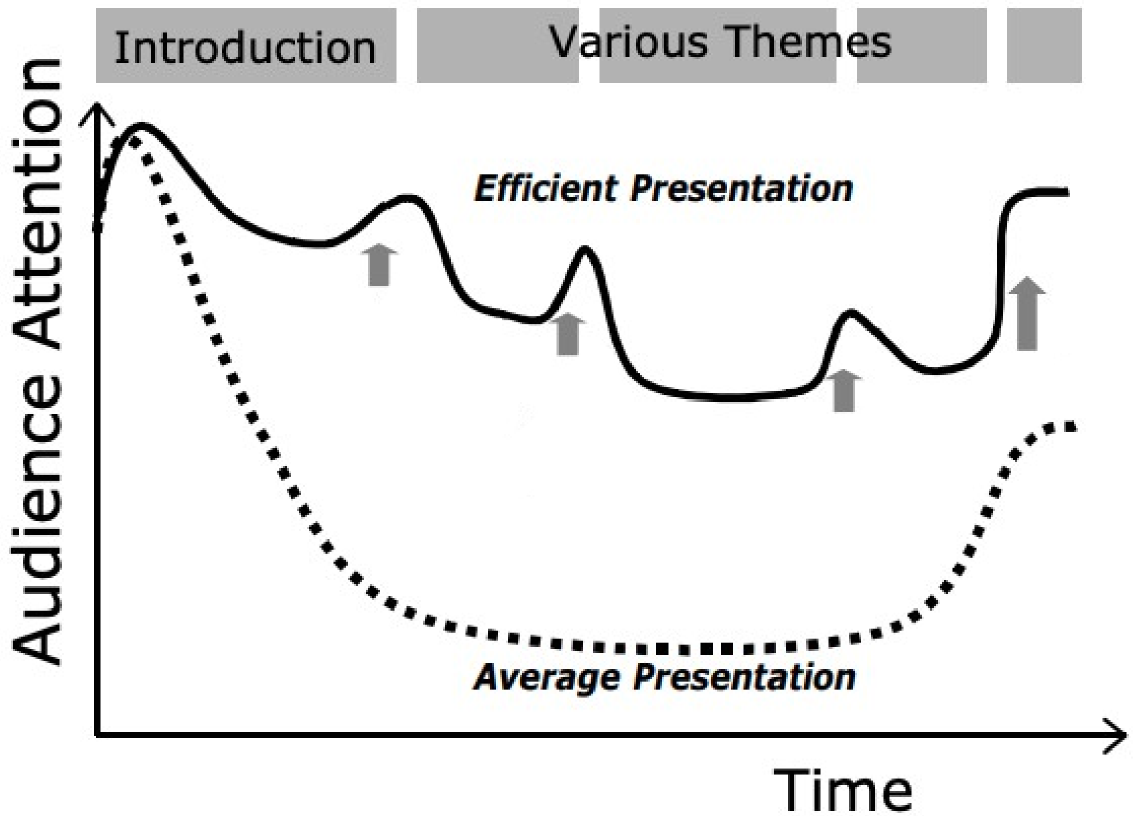


Figure 2 Ideal attention curve of an audience when the speaker divides his talk in recognizable parts, each summarized by intermediate conclusions. If people loose their attention for some reason, they can easily catch up with the speaker in one of his intermediate summaries. The big advantage of this approach is that every important item is said several times. Repeating the essentials is the key to getting your message across

1. **Emphasize the beginning and end**
2. **Help the audience follow along in the middle, assuming their attention span is at its worse**

Giving a good talk means understanding an audience

Baby the audience!

Like a baby, an audience:

- Sleeps often
- Has a short attention span
- Is easily confused
- Is easily overwhelmed
- Needs to be spoon-fed
- Needs to be told things over and over
- Depends on help from an adult

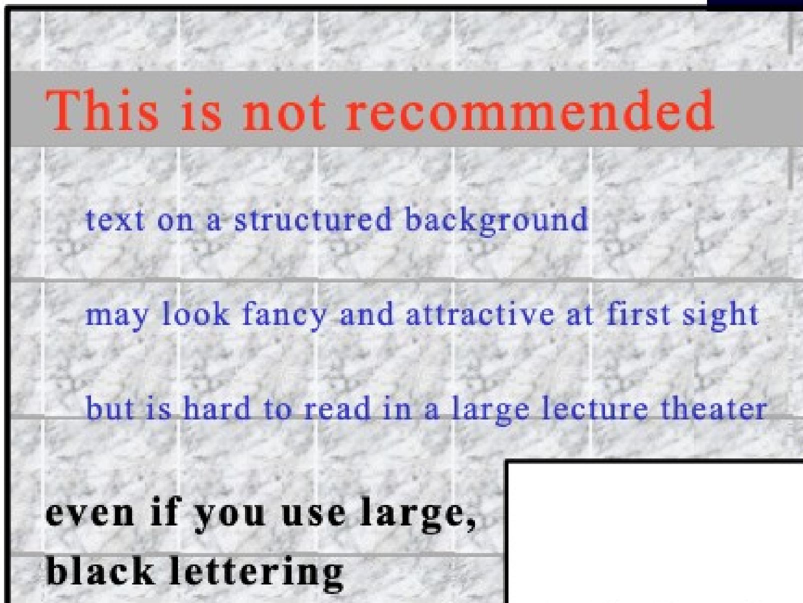
How to make your talk with the audience (baby) in mind

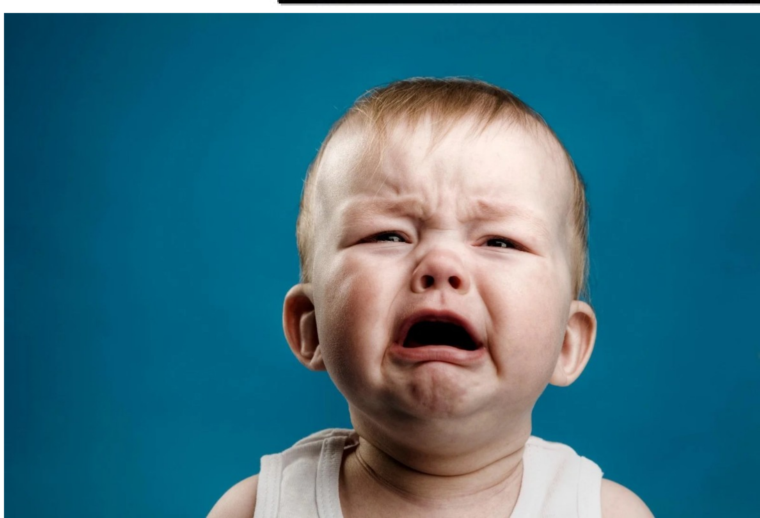
There are two complementary elements to preparing a talk:

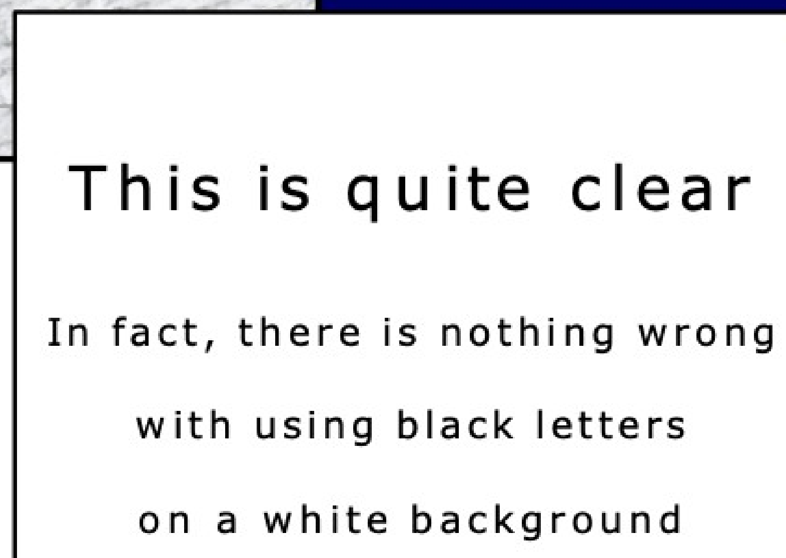
1. Preparing your slides
2. Preparing your delivery

There are opportunities to baby the audience at both steps!


This is acceptable
bright yellow or white
on the darkest possible blue
be careful with other colours
grey, green, orange, brown, blue
are much less clear


This is not recommended
text on a structured background
may look fancy and attractive at first sight
but is hard to read in a large lecture theater
even if you use large,
black lettering




This is quite clear
In fact, there is nothing wrong
with using black letters
on a white background

- The classic structure of oral presentations:

  - **Say what you are going to say** (Introduction)

  - **Say it** (the body of the talk)

  - **Say what you said** (wrap-up)

The classic structure of oral presentations:

Possible order 1:

- General observation in nature
- Unknown question or problem
- The solution: details of organism or system
- Objectives statement

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- Methods
- Results 1
- Results 2
- Interpretation and answer to question
- Future directions

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The classic structure of oral presentations:

Possible order 1:

- General observation in nature
- Unknown question or problem
- The solution: details of organism or system
- Objectives statement
- Methods 1
- Methods 2
- Results 1
- Results 2
- Interpretation and answer to question
- Future directions
- Summary and conclusion statement

Possible order 2:

- General observation in nature
- Unknown question or problem
- The solution: details of organism or system
- Objectives statement
- Methods: details of organism or system
- Methods: analysis 1
- Results analysis 1
- Methods: analysis 2
- Results analysis 2
- Interpretation and answer to question
- Summary and conclusion statement

Introduction: Slide preparation

Give background information:

- Start with an interesting observation in nature **(remember the audience's attention is at its highest during your first slide)**
- Gradually lead us to the knowledge gap
- Convince the audience that this gap is important
- Don't give background information unless relevant
- Do not assume basic information about your organism or system is common knowledge

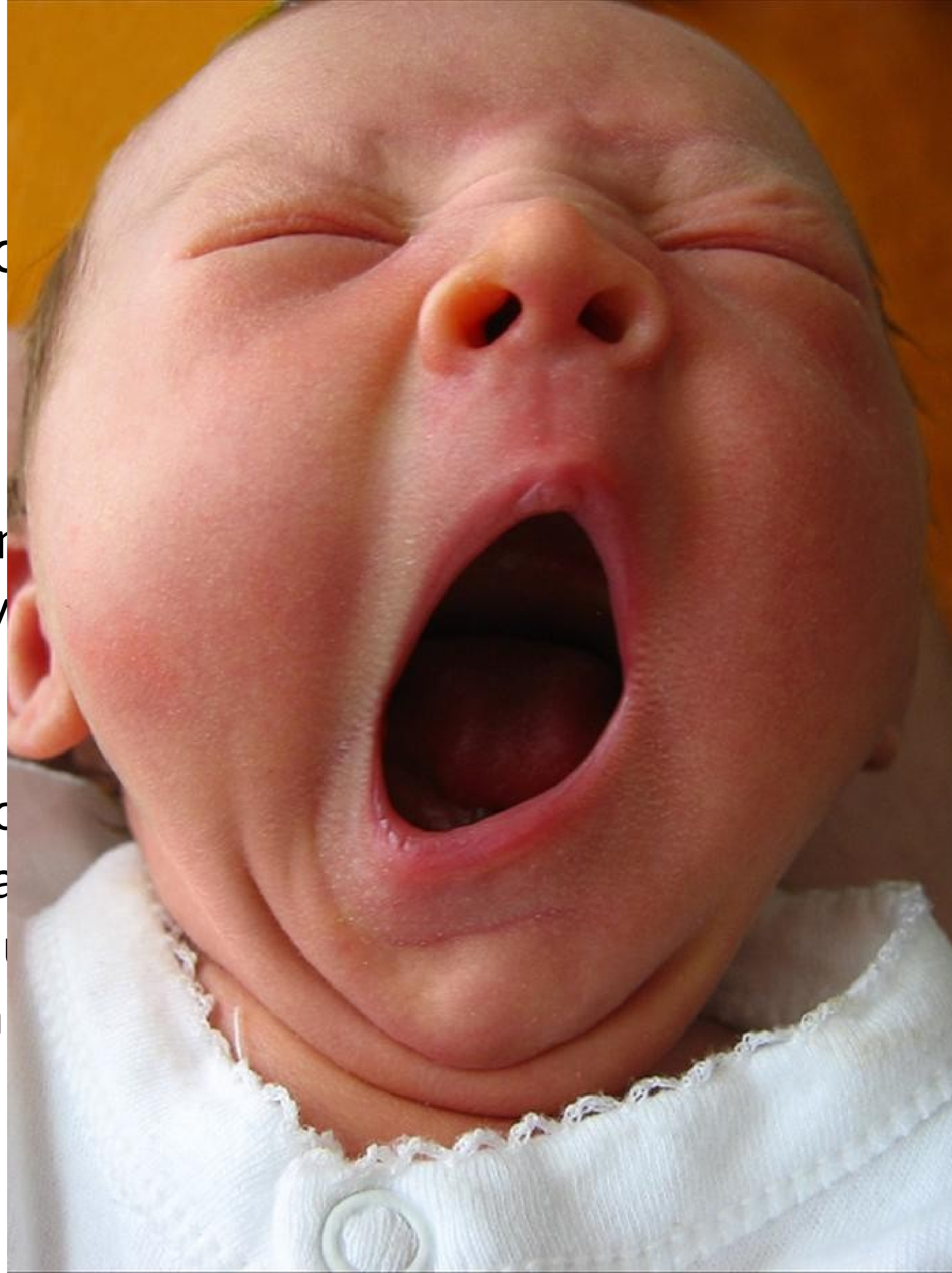
Example slide for Intro:

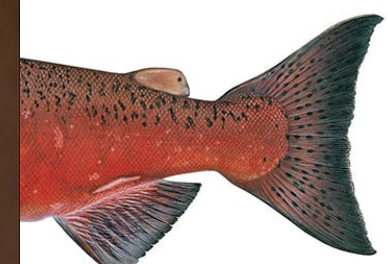
Chinook salmon *(Oncorhynchus tshawytscha)*

- Salmon spawn in freshwater
- Some individuals stay in the estuary for a period of time
- Others immediately leave to the ocean
- This may have different consequences on their growth rates

Example sl

Chinc *cha)*





- Spawn
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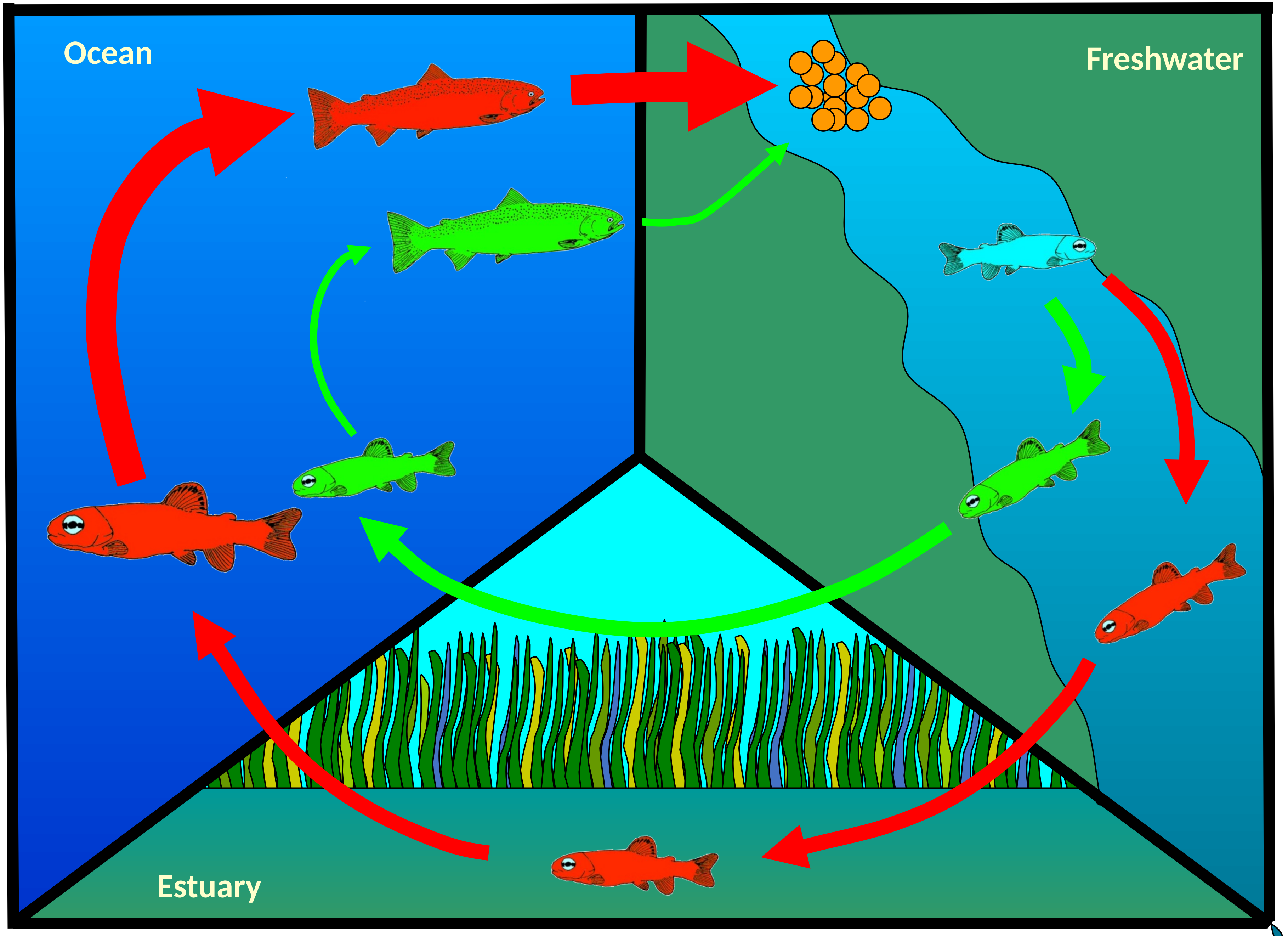
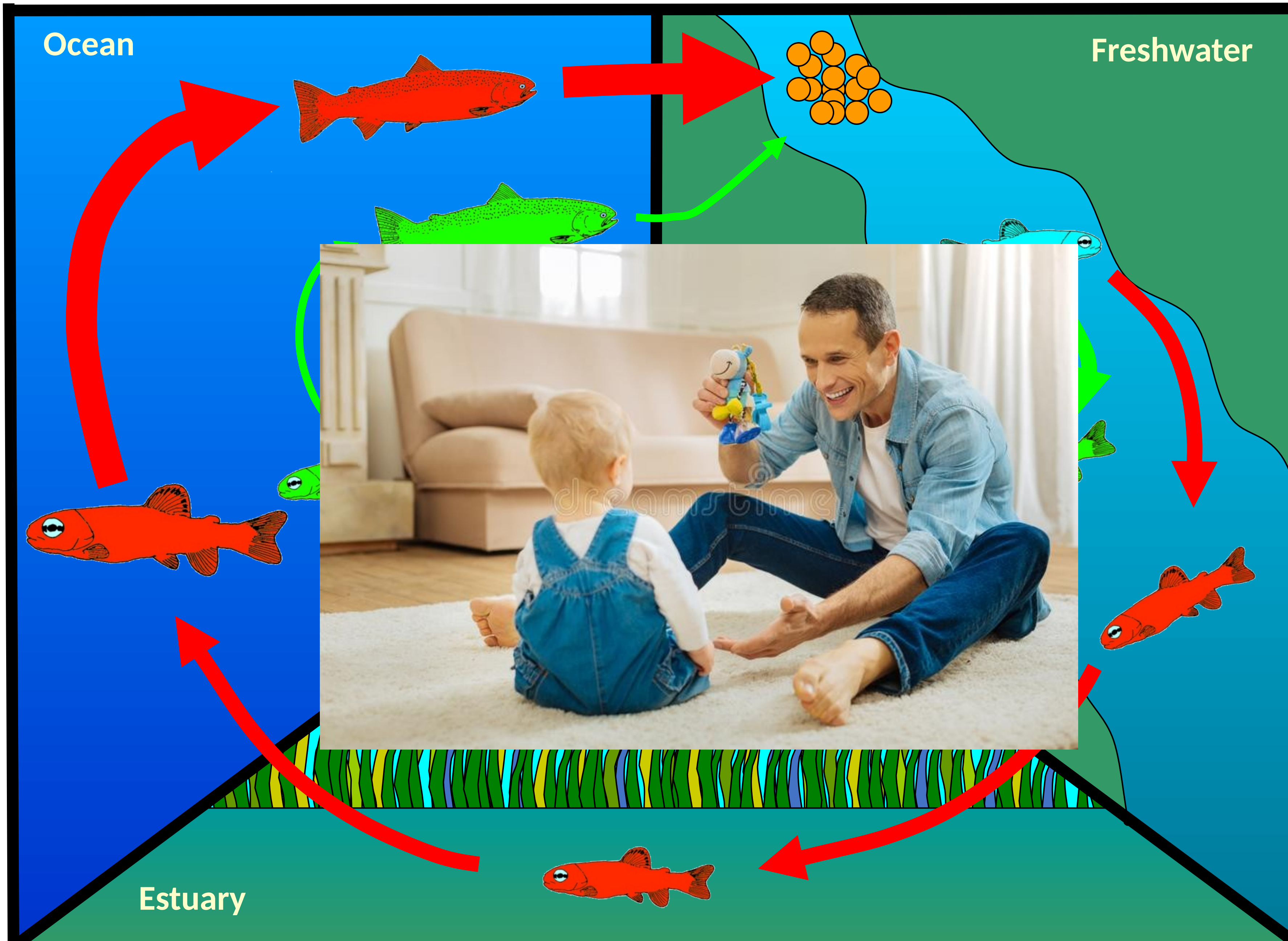

Ocean
Freshwater
Estuary


Ocean
Freshwater
Estuary

Introduction: Slide preparation

Objectives statement:

- SPOON FEED
- Should be explicit and clear, like in your papers
- Including hypotheses or predictions can help

The function of the objectives statement is to tell the audience what you are going to say.

It sets up expectations for the rest of the talk

QUESTION:

Does sunlight and cloud cover effect the duration of otter dives?

HYPOTHESIS:

On cloudy days, otters have shorter dives because plankton and prey are near the surface

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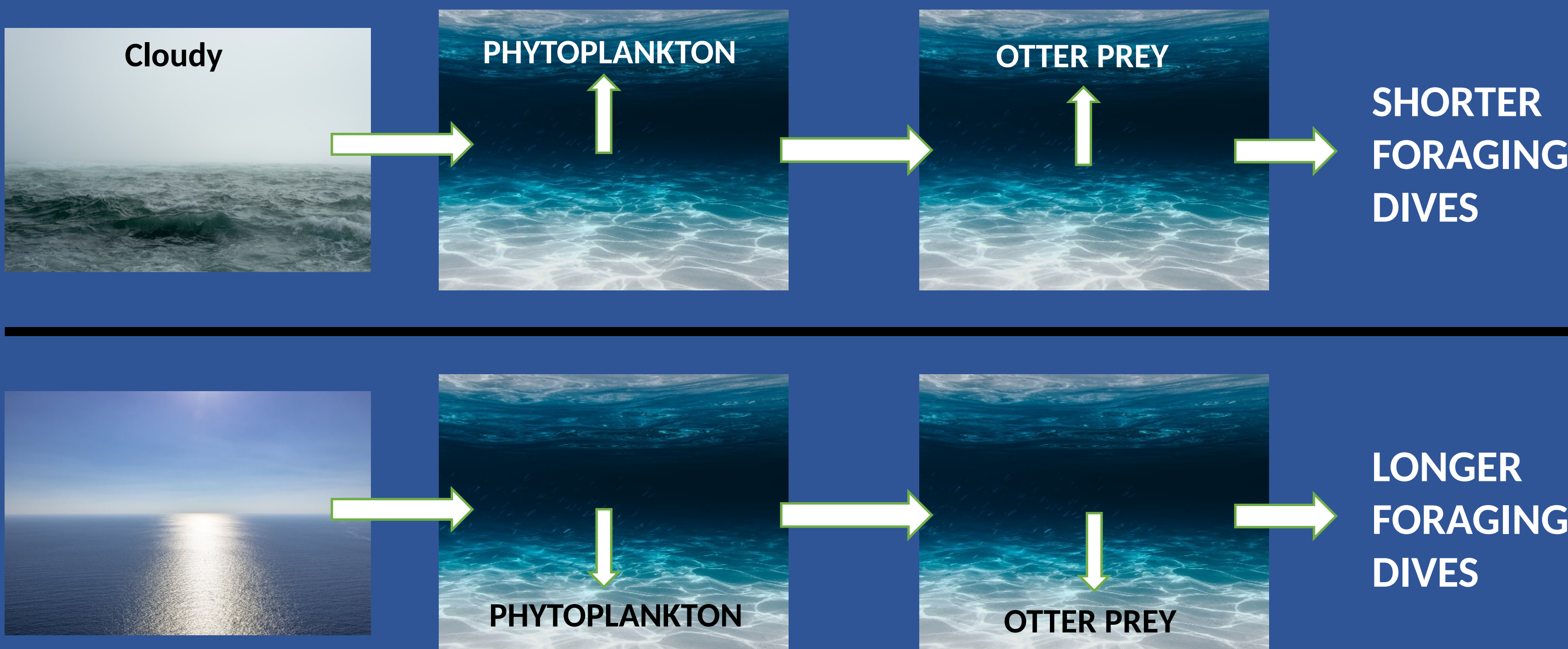
THE HYPOTHESIS:



Introduction: **Delivery tips**

Just as the audience is the most alert, **you are the most nervous.**

- Learn strategies to manage your nerves
- Practice the Introduction **the most** of any section

Form a **connection** with your audience at this stage

- Find common ground
- Invoke audience's emotions or sympathy
- Be extra enthusiastic and use eye contact

Methods: Slide preparation

Help us understand your approach:

- Methods can be hard to understand
- Break Methods down into steps
- Find simple ways to explain complicated analyses
- Use photographs and schematics

For example:

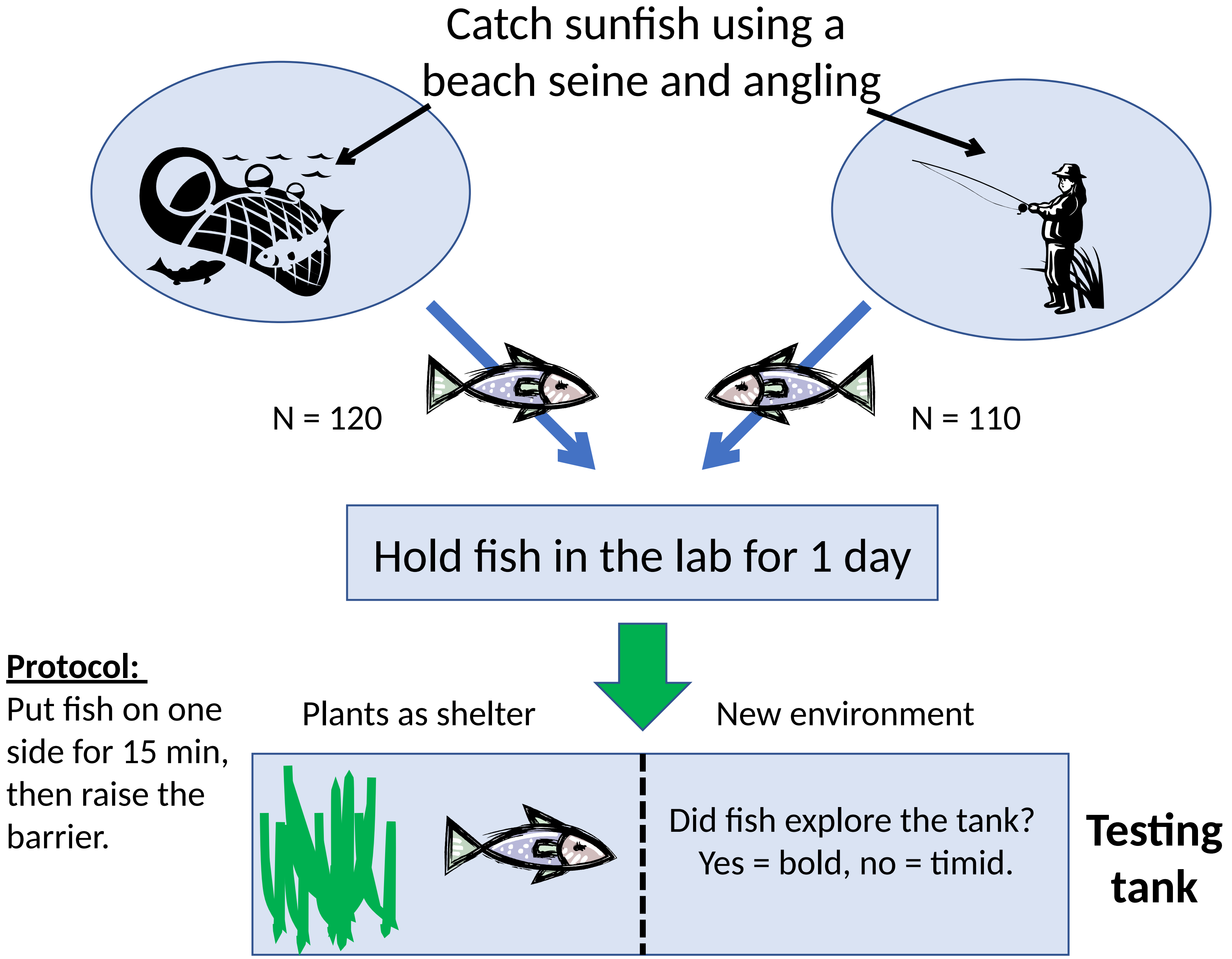
How could we convey the methods for this study?

Research Question:

Is fish behavior in captivity affected by whether they were captured by seining or hook-and-line?

Methods

- Fish were caught via hook and line (n=110) and beach seine (n=120)
- Fish were acclimated in an aquarium for 24 hours
- Plants were included in the shelter for protection
- A mesh wall was used to separate the fish from an empty area.
- The barrier was removed after 15 minutes and the fish was free to explore the empty area.
- Are you actually still reading this? If so, you’re probably not listening to me talk.


Catch sunfish using a beach seine and angling
N = 120
N = 110
Hold fish in the lab for 1 day
Protocol:
Put fish on one side for 15 min, then raise the barrier.
Plants as shelter
New environment
Did fish explore the tank?
Yes = bold, no = timid.
Testing tank

Methods: **Delivery tips**

Go slowly. If we don’t understand Methods, we can’t understand Results!

Results: Slide preparation

The moment we've all been waiting for.

- Remember you've led us to expect certain results based on your Objectives and Methods
- Choose the packaging wisely (text, table or figure)

<u>Avoid tables</u> – the audience gets lost and stops paying attention to you.

depth (m)	afternoon	evening	night	dawn	morning	Grand Total
10	203.7	180.9	189.6	175.4	214.6	184.5
25	198.3	210.2	213.6	202.2	185.7	200.6
50	223.3	231.9	233.6	225.5	226.4	228.8
70	262.0	255.9	261.6	260.4	259.3	259.9
total	218.2	213.6	211.2	206.4	208.8	211.4

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depth (m)	afternoon	evening	night	dawn	morning	Grand Total
10	203.7	180.9	189.6	175.4	214.6	184.5
25	198.3	210.2	213.6	202.2	185.7	200.6
50	223.3	231.9	233.6	225.5	226.4	228.8
70	262.0	255.9	261.6	260.4	259.3	259.9
total	218.2	213.6	211.2	206.4	208.8	211.4

If you must show tables:

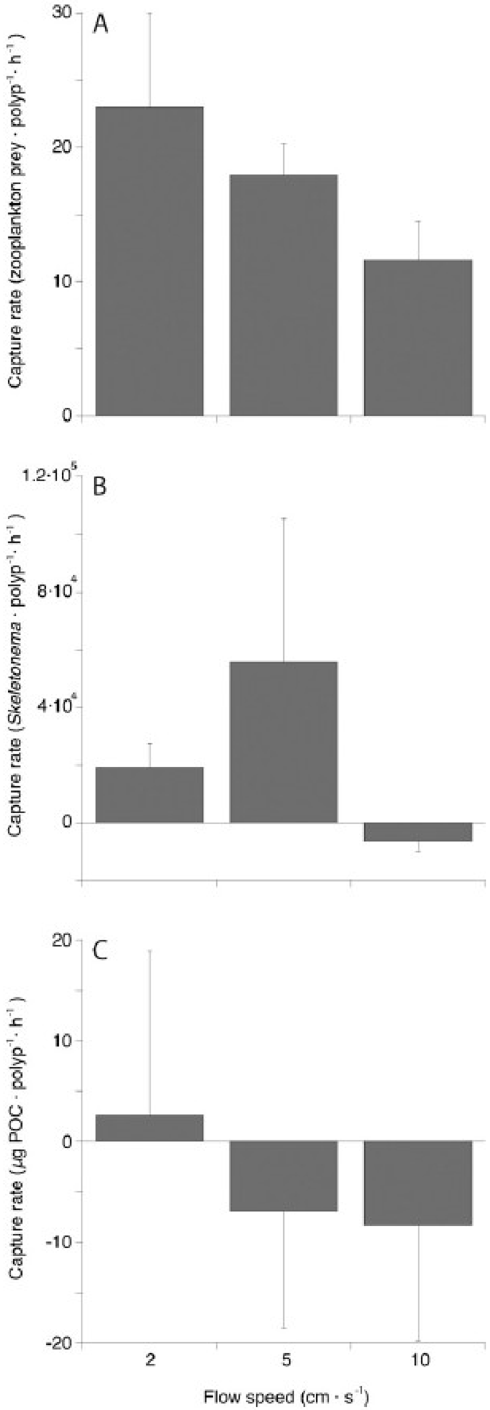
- Highlight pertinent results
- Make fonts large enough to read
- Avoid excessive decimal places
- Take the time to explain the table

Figures:

Figures are the essence of a talk, but must be done wisely!

- Only displayed for a short time.
- MUST be easily digestible!
- Must be visible from back of room

Results and figures in a talk will differ in style from those you present in your paper!



Contents lists available at ScienceDirect

Journal of Experimental Marine Biology and Ecology

journal homepage: www.elsevier.com/locate/jembe

The effect of flow speed and food size on the capture efficiency and feeding behaviour of the cold-water coral *Lophelia pertusa*

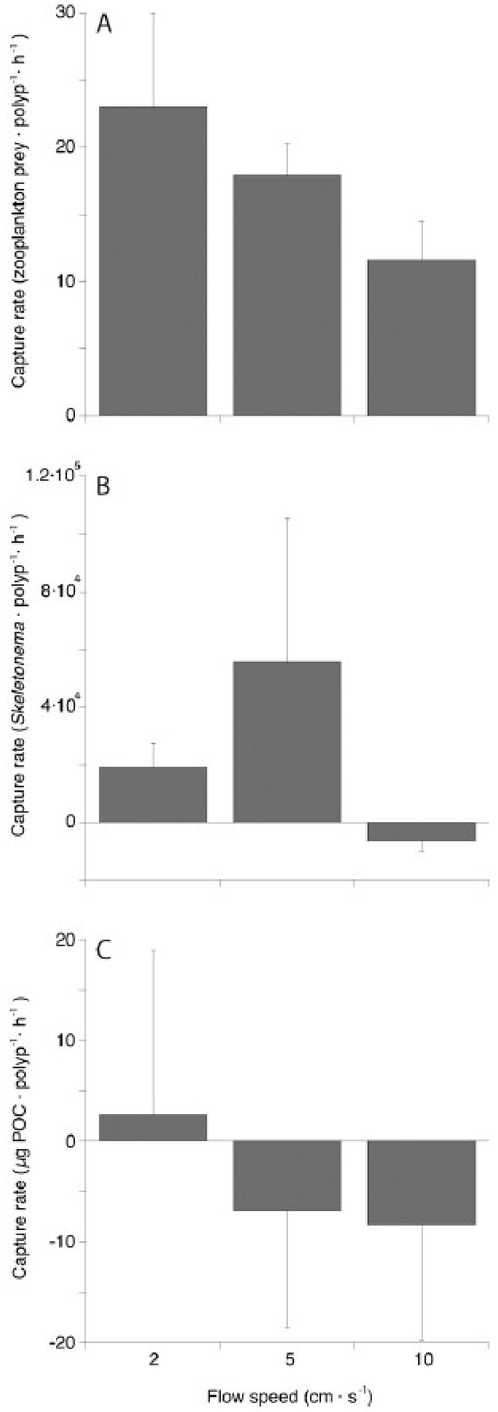
Covadonga Orejas [a,1,b,*,‡], Andrea Gori [b,c,d,2,‡], Cecilia Rad-Menéndez [e], Kim S. Last [e], Andrew J. Davies [f], Christine M. Beveridge [e], Daniel Sadd [f], Konstadinos Kiriakoulakis [g], Ursula Witte [h], John Murray Roberts [c,1,i]

Fig. 4. Capture rates (average ± SD) for *Lophelia pertusa* under three different flow speed (2, 5 and 10 cm/s) for the different prey items. A: Zooplankton capture rate (zooplankton prey per polyp per hour); B: algae capture rate (*Skeletonema* per polyp per hour); C: particulate organic carbon capture rate (μg POC per polyp per hour).

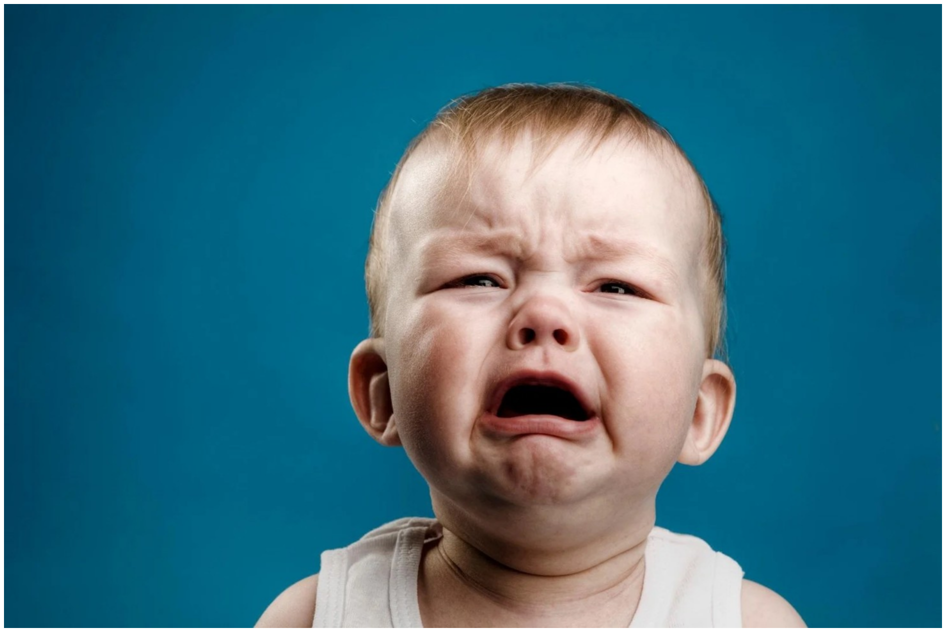
Basic example – little change at all from the paper

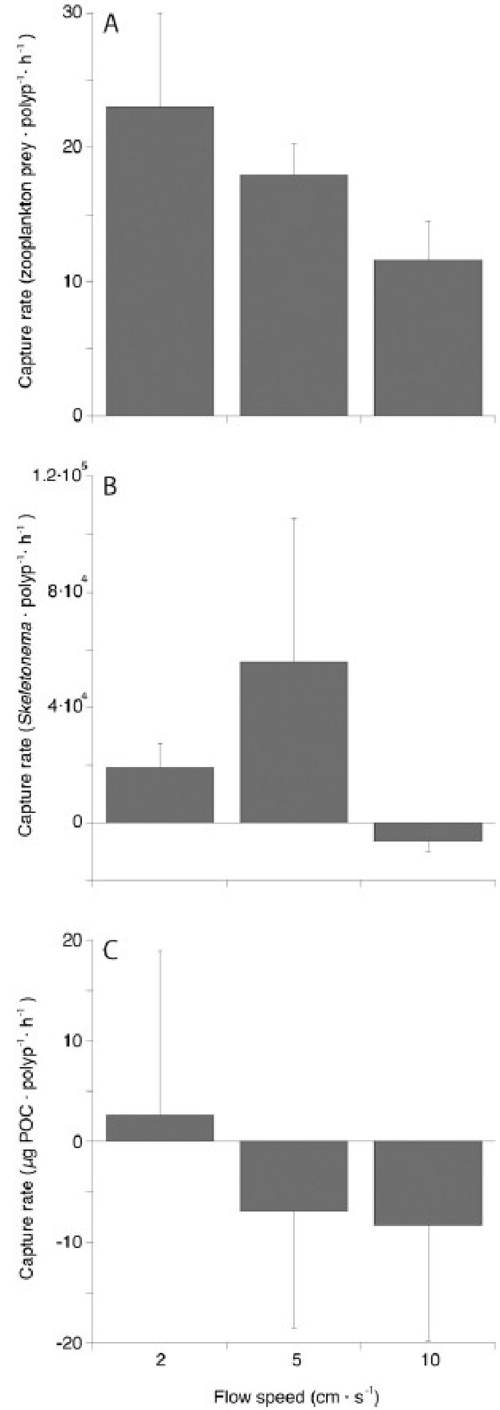
Results:

Capture rates of three different prey types across different flow speeds



Basic example – little change at all from the paper



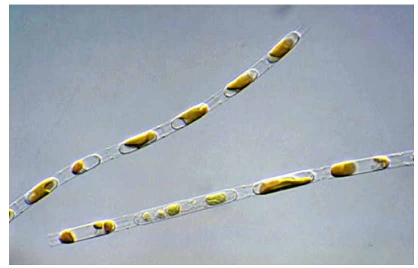


Better example:

Results

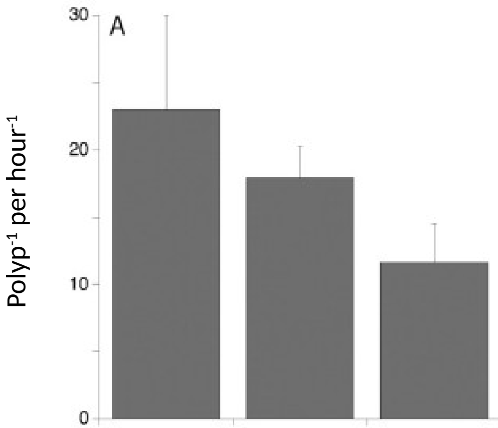
How does flow speed affect capture rates for:

zooplankton



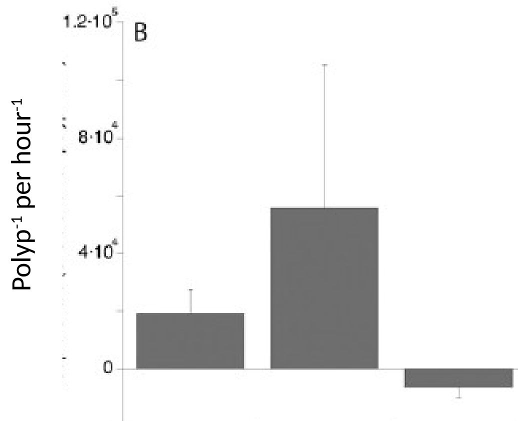
algae

POC



2 5 10

Flow Speed
(m/s)



2 5 10

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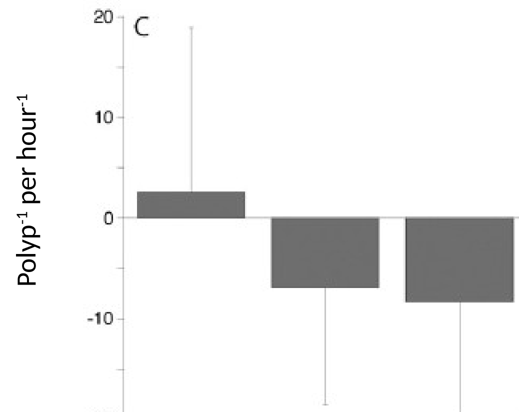


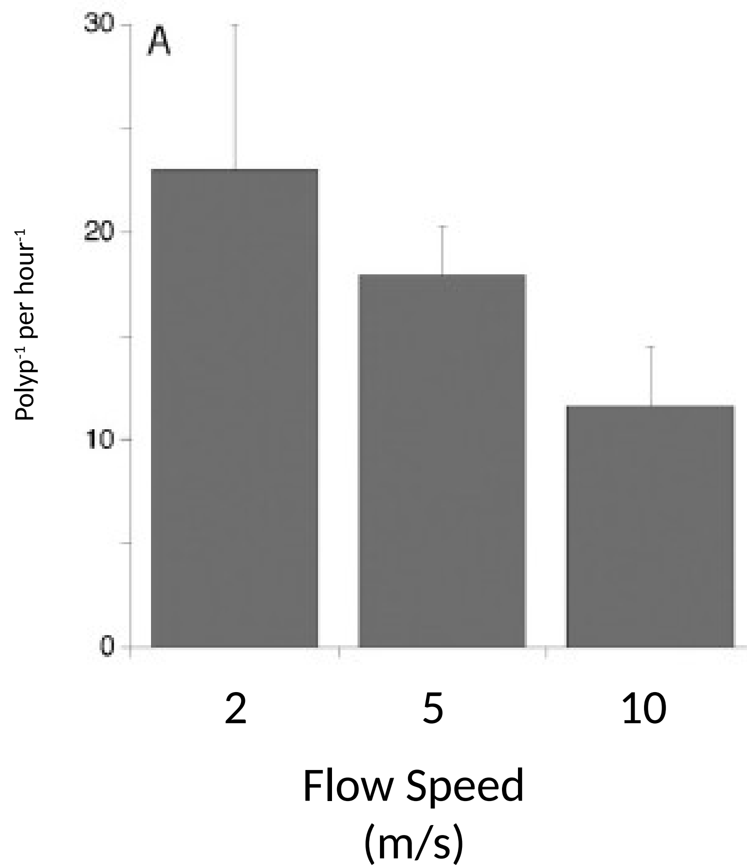
2 5 10

Flow Speed
(m/s)

Results

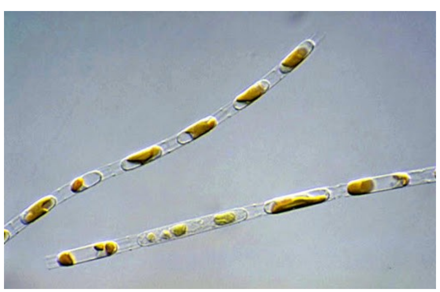
How does flow speed affect capture rates?

zooplankton



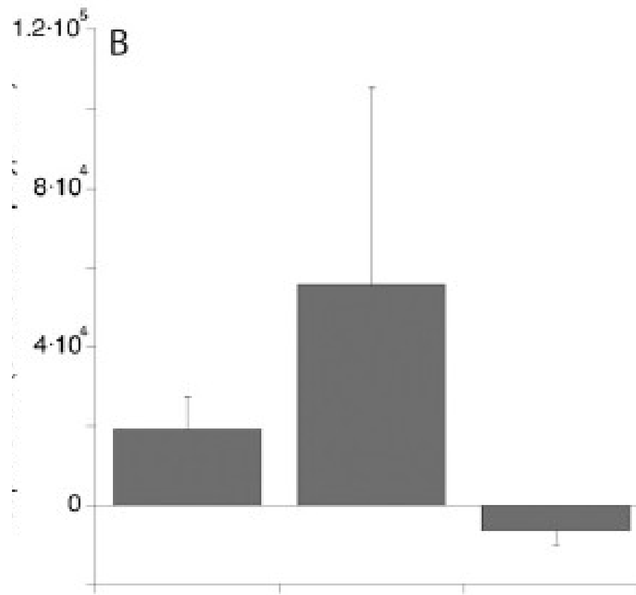
Results

How does flow speed affect capture rates?



algae

Polyp^{-1} per hour^{-1}



2 5 10

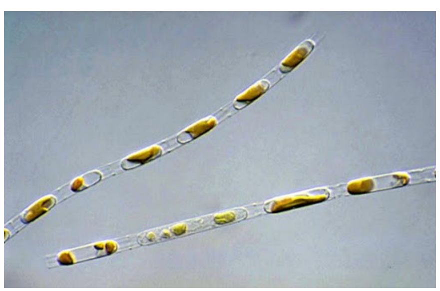
Flow Speed
(m/s)

Results: **Slide preparation tips**

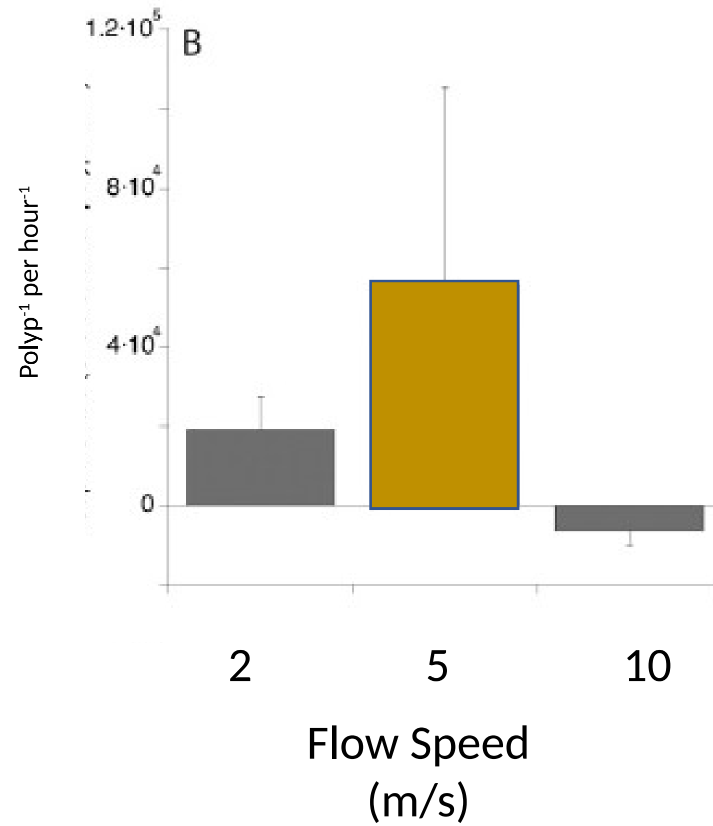
- **Use color wisely**
- **Don’t overwhelm the reader with too many results**
- **Balance graphics, photographs and text/numbers**
- **Balance white space**
- **Don’t be afraid to be “Captain Obvious”**

Results

Lophelia consume algae best at intermediate flow speeds



algae



Results: **Delivery tips**

When first showing a figure, say: what data are shown, what the x-axis is, and what the y-axis is.

- Yes we can read it, but people learn differently. Some learn visually and some learn with sound.
- Figures may show a lot of information and audience's initial response is deer-in-headlights. Direct our eyes by pointing out the axes first.
- After walking us through the figure, *then* give us the take home message when we are ready to understand it
- Give us a minute or more to understand figure: don't flip to next slide too soon

Interpretation: Slide preparation

You've just shown us hard data, **now make sense of it.**

- Why do you think you got these results (biologically)?
- Opportunity to point out other interesting things that you weren't expecting
- Lots of freedom in this section, but don't go overboard. You may be close to times-up!

Giving a good talk means understanding an audience

1. Emphasize the beginning and end

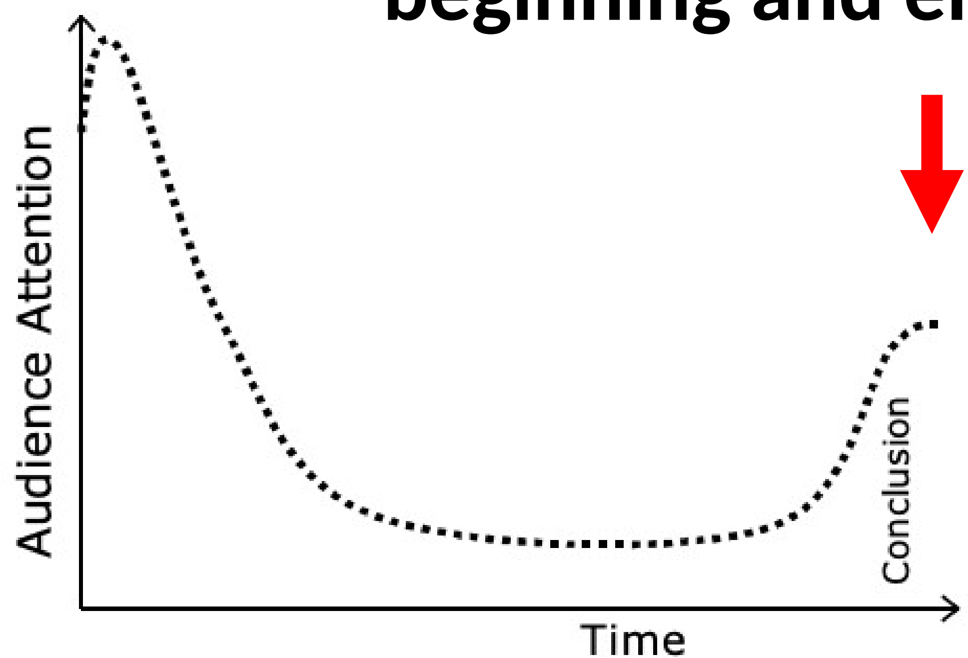


Figure 1 Typical attention the audience pays to an average presentation

Conclusion: Slide preparation

Give us the take-home!

- We will remember the end best of all
- **Remind us what you told us: give us a summary of the question you asked and the answer you found**
- End on a high-note: do not end with apologies or caveats
- Keep it short and sweet
- "To summarize, I asked…I found…This means…"

The ending: **Delivery tips**

Stick the landing.

- Audience is waiting for the talk to be over to clap. How do they know it's over?
- Make it obvious:
  - "With that, I'll take questions. Thank you."
  - Or your own version of a send-off
- Don't have to read Acknowledgements off, can simply display them

Great, you've made your slides. Now what?

Time to **Practice** and **Edit!**

1. **Practice**
   - *Stay within 10 minutes*. Don't rob time from your audience and the next speaker
   - Especially rehearse the beginning
   - Practicing cures a case of the "ums"
   - Please don't read directly from notes
2. **Edit**
   - Resist urge to add more: we can only remember so much in 10 minutes
   - If you need to cut slides, cut from the middle

Assignments (we’re almost done!)

1. Peer edits are due Friday at 11:49 pm

2. Work on your talk. Talks begin Mar 13th and order will be random
3. Final papers due Mar 13th